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MODALITIES IN THE *SURVEY* SYSTEM OF STRICT IMPLICATION

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I. Introduction. Professor C. I. Lewis, in Lewis and Langford's *Symbolic logic*, designates the system (S2) determined by the postulates used in Chapter VI—namely, 11.1–.7 (B1–7) and

19.01 (B8) $\Diamond(pq) \prec \Diamond p$ —

as *the* system of strict implication.[1] For certain reasons,[2] he prefers it to either the earlier system (S3) determined by the stronger set of postulates of his *Survey of symbolic logic*, as emended, namely, A1–7 and

A8 $p \prec q . \prec . \sim\Diamond q \prec \sim\Diamond p,$

or the system (S1) determined by the weaker set of postulates B1–7 or A1–7.

But Lewis and others, following O. Becker,[3] have also given consideration to systems which contain some additional principle effecting the reduction of complex modalities to simpler ones. Notable are the system (S4) determined by B1–7 plus

C10 $\sim\Diamond\sim p \prec \sim\Diamond\sim\sim\Diamond\sim p,$

which includes (is stronger than) S3; and the system (S5) determined by B1–7 plus

C11 $\Diamond p \prec \sim\Diamond\sim\Diamond p,$

which includes S4.

It seems worth while to investigate further the system of the *Survey* (emended), S3, which is intermediate between S2 and S4. This is the purpose of the present paper. We first prove some additional theorems in S2 and S3. These enable us to reduce all the complex modalities in S3 to a finite number, viz. 42; and it is shown that no further reduction is possible. Finally, several systems which include S3 are considered.

We take the following as the *postulates* for S3:

11.1 $pq \prec qp$
11.2 $pq \prec p$
11.3 $p \prec pp$
11.4 $(pq)r \prec p(qr)$
11.6 $p \prec q . q \prec r . \prec . p \prec r$
11.7 $p . p \prec q . \prec . q$
30.1 $p \prec q . \prec . \sim\Diamond q \prec \sim\Diamond p$

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[1] We shall leave out of account the existence postulate 20.01 (B9) of *Symbolic logic*, which may be considered either as present in all of the systems to be discussed or as absent in all of them.

[2] Op. cit., pp. 177f., 496.

[3] *Zur Logik der Modalitäten, Jahrbuch für Philosophie und phänomenologische Forschung*, vol. 11 (1930), pp. 497–548.

The primitive ideas, rules of procedure (Substitution—including replacement of equivalents, Adjunction, and Inference) and definitions (11.01.02.03, 14.01.02, 17.01) are as given in *Symbolic logic*, Chapter VI.

McKinsey[4] has shown that from 11.02.03.1.2.3.6 follows as theorem

11.5 $p \prec \sim\sim p$.

From 11.1–.7, the postulates given in Section 1 of Chapter VI (determining S1), follow all the theorems of Sections 1 through 4, numbered 12.1–18.92. We may therefore refer to any of these theorems, by number as there given, as theorems of S3, or of any system including S1.

S3 obviously gives as a theorem

19.01 $\Diamond(pq) \prec \Diamond p$. [11.2, 30.1, 12.44]

From 11.1–.7, 19.01—the postulates of S2—follow in addition the theorems of Chapter VI, Section 5, numbered 19.02–.92; likewise the theorems of E. V. Huntington's *Postulates for assertion, conjunction, negation, and equality*,[5] an axiomatization of S2 on a different base (see especially Section 9).

11.1, 12.17, 11.3, 12.5, 11.5.6, 18.41, 30.1 give the postulates of Chapter V of the *Survey* as emended: 1.1–.7, 2.2 (replacing 1.8); i.e., A1–8.[6]

II. Theorems of S2. First, the additional *theorems of S2* that we shall need—of which 22.4.41.42.7 are in fact *theorems of S1*:

21.9 $p \prec q . = . p = pq$ (Cf. Huntington, Th. 97 and note 1, p. 2.)

[19.62] $p \prec pq . \prec . p \prec p . p \prec q .$

[4] J. C. C. McKinsey, *A reduction in number of the postulates for C. I. Lewis's system of strict implication*, *Bulletin of the American Mathematical Society*, vol. 40 (1934), pp. 425–427.

[5] *Proceedings of the American Academy of Arts and Sciences*, vol. 72 (1937), pp. 1–44; cited in this paper as "Huntington."

[6] An alternative set of postulates for S3 is obtained by adding to the postulates for S2 the principle

31.12 $p = q . \prec . \sim\Diamond\sim p = \sim\Diamond\sim q,$

proved below as a theorem of S3. 30.1 (A8) can then be proved, as follows:

[19.81] $(31.12\, pq/q) = : p = pq . \prec . \sim\Diamond\sim p = \sim\Diamond\sim p \sim\Diamond\sim q$ (1)

[21.9, $(21.9 \sim\Diamond\sim p/p, \sim\Diamond\sim q/q)$] (1) $= : p \prec q . \prec . \sim\Diamond\sim p \prec \sim\Diamond\sim q$ (2)

[12.44, 12.3, $(12.3\; q/p)$] $\{(2) \sim q/p, \sim p/q\}$ = A8

The same effect would of course be produced by adding any of the principles

$$p = q . \prec . \Diamond p = \Diamond q, \quad p = q . \prec . \sim\Diamond q \prec \sim\Diamond p,$$

etc., instead of 31.12.

This postulate set provides a sort of analysis of S3 in comparison with S2, making clearer just what S3 involves. 31.12, it seems to me, is as intuitively evident, is as acceptable a principle of deduction, as any of the postulates of S2 (or at least, as any of those of the second degree, i.e., B6–8; see note 11 below). It may profitably be compared with the principle of replacement of equivalents, according to which, for example, if $P = Q$, then (since $\sim\Diamond\sim P = \sim\Diamond\sim P$) $\sim\Diamond\sim P = \sim\Diamond\sim Q$.

[11.2] $\prec\!\!3 \,.\, p \prec\!\!3\, p \,.$
[19.51] $\prec\!\!3 \,.\, pq \prec\!\!3\, p$ (1)
[16.33] (1) $= : p \prec\!\!3\, pq \,.\, \prec\!\!3 \,.\, p \prec\!\!3\, pq \,.\, pq \prec\!\!3\, p$ (2)
[11.2] $p \prec\!\!3\, pq \,.\, pq \prec\!\!3\, p \,.\, \prec\!\!3 \,.\, p \prec\!\!3\, pq$ (3)
[11.03] (2) (3) $= : p \prec\!\!3\, pq \,.\, = \,.\, p \prec\!\!3\, pq \,.\, pq \prec\!\!3\, p$ (4)
[16.33, 11.03] (4) = QED.

22.1 $p \prec\!\!3 \sim\Diamond\sim q \,.\, \prec\!\!3 \,.\, p \prec\!\!3\, q$
[21.9] (18.42 q/p) $= : \sim\Diamond\sim q = \,.\, \sim\Diamond\sim q \,.\, q$ (1)
[19.62] $p \prec\!\!3 \,.\, \sim\Diamond\sim q \,.\, q : \prec\!\!3 : p \prec\!\!3 \sim\Diamond\sim q \,.\, p \prec\!\!3\, q :$
[12.17] $\prec\!\!3 : p \prec\!\!3\, q$ (2)
[(1)] (2) = QED.

22.11 $p \prec\!\!3 \sim\Diamond\Diamond\sim q \,.\, \prec\!\!3 \,.\, p \prec\!\!3\, q$
[11.6] (22.1 $\sim\Diamond\sim q/q$) (22.1) $. \prec\!\!3 : p \prec\!\!3 \sim\Diamond\sim\sim\Diamond\sim q \,.\, \prec\!\!3 \,.\, p \prec\!\!3\, q$ (1)
[12.3] (1) = QED.

22.2 $p \prec\!\!3 \,.\, q \prec\!\!3\, r : \prec\!\!3 \,.\, pq \prec\!\!3\, r$ [22.1] $p \prec\!\!3 \,.\, \sim\Diamond\sim(q \supset r) : \prec\!\!3 : p \prec\!\!3 \,.\, q \supset r$ (1)
[18.7, 14.26] (1) = QED.

22.3 If $P \prec\!\!3\, Q$ is established, then $\sim\Diamond Q \prec\!\!3 \sim\Diamond P$ may be asserted. (Cf. "Becker's Rule."[7])
[Hypothesis] $P \prec\!\!3\, Q$ (1)
[21.9] (1) $= : P = PQ$ (2)
[(2)] (19.13 P/p, Q/q) $= : \Diamond P \prec\!\!3 \Diamond Q$ (3)
[12.44] (3) $= : \sim\Diamond Q \prec\!\!3 \sim\Diamond P$

22.31 If $P \prec\!\!3\, Q$ is established, then $\sim\Diamond\Diamond Q \prec\!\!3 \sim\Diamond\Diamond P$ may be asserted.
[12.44] (22.3) $= :$ If $P \prec\!\!3\, Q$ is established, then $\Diamond P \prec\!\!3 \Diamond Q$ may be asserted. (1)
[(1), (22.3 $\Diamond P/P$, $\Diamond Q/Q$)] QED.

22.4 $\sim\Diamond\Diamond p \prec\!\!3 \sim\Diamond p$ [12.44] (18.4 $\Diamond p/p$) = QED.

22.41 $\sim\Diamond\Diamond\sim\Diamond p \prec\!\!3 \sim\Diamond\sim\Diamond p$ [22.4]

22.42 $\sim\Diamond\Diamond\sim\Diamond\Diamond p \prec\!\!3 \sim\Diamond\sim\Diamond\Diamond p$ [22.4]

22.5 $\sim\Diamond\sim\Diamond p \prec\!\!3 \sim\Diamond\sim\Diamond\Diamond p$ [22.3, 22.4]

22.55 $\sim\Diamond\sim\Diamond\sim\Diamond\Diamond p \prec\!\!3 \sim\Diamond\sim\Diamond\sim\Diamond p$ [22.3, 22.5]

22.6 $\sim\Diamond\Diamond\sim\Diamond p \prec\!\!3 \sim\Diamond\Diamond\sim\Diamond\Diamond p$ [22.31, 22.4]

22.65 $\sim\Diamond\sim\Diamond\Diamond\sim\Diamond\Diamond p \prec\!\!3 \sim\Diamond\sim\Diamond\Diamond\sim\Diamond p$ [22.3, 22.6]

22.7 $p = q \,.\, = \,.\, \sim p = \sim q$
[12.15] (11.03) $= : p = q \,.\, = \,.\, q \prec\!\!3\, p \,.\, p \prec\!\!3\, q$ (1)
[(12.44 q/p, p/q), 12.44] (1) $= : p = q \,.\, = \,.\, \sim p \prec\!\!3 \sim q \,.\, \sim q \prec\!\!3 \sim p$ (2)
[11.03] (2) = QED.

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[7] C. W. Churchman, *On finite and infinite modal systems*, this JOURNAL, vol. 3 (1938), pp. 77–82; see pp. 79f.

22.8 $\sim\Diamond\sim\Diamond(p \supset \sim\Diamond\sim p)$

[14.2] $\sim\Diamond\sim p \supset \Diamond\sim\Diamond\sim p = \sim(\sim\Diamond\sim p) \vee \Diamond\sim\Diamond\sim p$

[12.3] $= \Diamond\sim p \vee \Diamond\sim\Diamond\sim p$

[19.82] $= \Diamond(\sim p \vee \sim\Diamond\sim p)$

[14.2] $= \Diamond(p \supset \sim\Diamond\sim p)$ (1)

[18.7] $(18.4\ \sim\Diamond\sim p/p) = : \sim\Diamond\sim(\sim\Diamond\sim p \supset \Diamond\sim\Diamond\sim p)$ (2)

[(1)] (2) = QED.

III. Theorems of S3. We proceed to the proof of *theorems of S3*.[8]

31.1 $p \prec q . \prec . \Diamond p \prec \Diamond q$ [12.44] (30.1) = QED. (*Survey*, 2.74)

31.11 $p \prec q . \prec . \sim\Diamond\sim p \prec \sim\Diamond\sim q$ (*Survey*, 2.73)

[12.44] $(30.1\ \sim q/p, \sim p/q)$ = QED.

31.12 $p = q . \prec . \sim\Diamond\sim p = \sim\Diamond\sim q$

[19.68] $(31.11)(31.11\ q/p, p/q) . \prec :$

$p \prec q . q \prec p . \prec . \sim\Diamond\sim p \prec \sim\Diamond\sim q . \sim\Diamond\sim q \prec \sim\Diamond\sim p$ (1)

[11.03] (1) = QED.

31.2 $\sim\Diamond p . \prec . \sim\Diamond q \prec \sim\Diamond p$ [11.6] $(19.74)(30.1) \prec$ QED.

31.21 $\sim\Diamond\sim p . \prec . \sim\Diamond\sim q \prec \sim\Diamond\sim p$ [31.2 $\sim p/p, \sim q/q$]

31.22 $p \prec q . \prec : r \prec s . \prec . p \prec q$

[18.7, (18.7 $r/p, s/q$)] $(31.21\ p \supset q/p, r \supset s/q)$ = QED.

31.3 $p \supset q . \prec . r \supset s : \prec : p \prec q . \prec . r \prec s$

[18.7, (18.7 $r/p, s/q$)] $(31.11\ p \supset q/p, r \supset s/q)$ = QED.

31.31 $p \prec q . \prec : q \prec r . \prec . p \prec r$

[31.3] $(15.9) \prec :. p \prec q . \prec : q \supset r . \prec . p \supset r :$

[31.3] $\prec : q \prec r . \prec . p \prec r$

31.32 $q \prec r . \prec : p \prec q . \prec . p \prec r$

[31.3] $(15.92) \prec :. q \prec r . \prec : p \supset q . \prec . p \supset r :$

[31.3] $\prec : p \prec q . \prec . p \prec r$

31.4 $pq \prec r . \prec : \sim\Diamond\sim p \prec . \sim\Diamond\sim q \prec \sim\Diamond\sim r$

[14.26, 18.7] $(31.11\ q \supset r/q) = :. pq \prec r : \prec : \sim\Diamond\sim p \prec . q \prec r$ (1)

[31.32] $(31.11\ q/p, r/q) \prec :.$

$\sim\Diamond\sim p \prec . q \prec r : \prec : \sim\Diamond\sim p \prec . \sim\Diamond\sim q \prec \sim\Diamond\sim r$ (2)

[11.6] $(1)(2) \prec$ QED.

31.5 $p \supset q . r \supset s . \prec . t \supset u : \prec :. p \prec q . \prec : r \prec s . \prec . t \prec u$

[18.7] $(31.4\ p \supset q/p, r \supset s/q, t \supset u/r)$ = QED.

31.51 $p \supset q . r \supset s . \prec . t \supset u : \prec : p \prec q . r \prec s . \prec . t \prec u$

[11.6] $(31.5)\ (22.2\ p \prec q/p, r \prec s/q, t \prec u/r) \prec$ QED.

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[8] Theorems 31.2–.32, 31.5–32.31 are contained in my doctoral thesis, ***Implication***, on file in the Harvard University Library.

31.52 $p \strictif q \cdot \strictif : p \strictif r \cdot \strictif \cdot p \strictif qr$
[16.8] $p \supset q \cdot p \supset r \cdot \strictif \cdot p \supset qr$ (1)
[31.5] (1) $\strictif$ QED.

31.53 $p \strictif r \cdot \strictif : q \strictif s \cdot \strictif \cdot pq \strictif rs$ [31.5] (16.85) $\strictif$ QED.

32.1 $\sim\Diamond\sim\sim\Diamond\sim q \strictif \cdot \sim\Diamond\sim p \strictif \sim\Diamond\sim\sim\Diamond\sim p$
[31.21] $\sim\Diamond\sim\sim\Diamond\sim q \strictif \cdot \sim\Diamond\sim p \strictif \sim\Diamond\sim\sim\Diamond\sim q$ (1)
[31.21] $\sim\Diamond\sim p \cdot \strictif \cdot \sim\Diamond\sim q \strictif \sim\Diamond\sim p.$
[31.11] $\strictif \cdot \sim\Diamond\sim\sim\Diamond\sim q \strictif \sim\Diamond\sim\sim\Diamond\sim p$ (2)
[31.52] (2) $\strictif :. \sim\Diamond\sim p \strictif \sim\Diamond\sim\sim\Diamond\sim q \cdot \strictif :$
$\sim\Diamond\sim p \strictif \cdot \sim\Diamond\sim\sim\Diamond\sim q \strictif \sim\Diamond\sim\sim\Diamond\sim p \cdot \sim\Diamond\sim\sim\Diamond\sim q$ (3)
[12.15] (11.7 $\sim\Diamond\sim\sim\Diamond\sim q/p, \sim\Diamond\sim\sim\Diamond\sim p/q) = :$
$\sim\Diamond\sim\sim\Diamond\sim q \strictif \sim\Diamond\sim\sim\Diamond\sim p \cdot \sim\Diamond\sim\sim\Diamond\sim q \cdot \strictif \cdot \sim\Diamond\sim\sim\Diamond\sim p$ (4)
[31.32] (4) $\strictif :.$
$\sim\Diamond\sim p \strictif \cdot \sim\Diamond\sim\sim\Diamond\sim q \strictif \sim\Diamond\sim\sim\Diamond\sim p \cdot \sim\Diamond\sim\sim\Diamond\sim q : \strictif :$
$\sim\Diamond\sim p \strictif \sim\Diamond\sim\sim\Diamond\sim p$ (5)
[12.78] (1)(3)(5) $\strictif$ QED.

32.2 $\sim\Diamond\sim\sim\Diamond\sim p \strictif \sim\Diamond\sim\sim\Diamond\sim\sim\Diamond\sim p$
[22.2] (32.1 $p/q, \sim\Diamond\sim p/p) \strictif :$
$\sim\Diamond\sim\sim\Diamond\sim p \cdot \sim\Diamond\sim\sim\Diamond\sim p \cdot \strictif \cdot \sim\Diamond\sim\sim\Diamond\sim\sim\Diamond\sim p$ (1)
[12.7] (1) $=$ QED.

32.21 $\sim\Diamond\sim\sim\Diamond\sim p = \sim\Diamond\sim\sim\Diamond\sim\sim\Diamond\sim p$
[11.03] (32.2)(18.42 $\sim\Diamond\sim\sim\Diamond\sim p/p) =$ QED.

32.22 $\Diamond\Diamond\Diamond p \strictif \Diamond\Diamond p$ [12.3] (32.2 $\sim p/p) = : \sim\Diamond\Diamond p \strictif \sim\Diamond\Diamond\Diamond p$(1)
[12.44] (1) $=$ QED.

32.23 $\Diamond\Diamond p = \Diamond\Diamond\Diamond p$ [11.03] (18.4 $\Diamond\Diamond p/p$)(32.22) $=$ QED.

32.3 $\sim\Diamond p \strictif \sim\Diamond\sim\Diamond\sim\Diamond p$
[31.2] $\sim\Diamond p \cdot \strictif \cdot \sim\Diamond q \strictif \sim\Diamond p.$
[30.1] $\strictif \cdot \sim\Diamond\sim\Diamond p \strictif \sim\Diamond\sim\Diamond q.$
[22.1] $\strictif \cdot \sim\Diamond\sim\Diamond p \strictif \Diamond q$ (1)
[18.14] $\{(1) \sim\Diamond p/q\} =$ QED.

32.31 $\sim\Diamond\sim\Diamond\sim\Diamond\sim\Diamond p = \sim\Diamond\sim\Diamond p$
[30.1] (32.3) $\strictif : \sim\Diamond\sim\Diamond\sim\Diamond\sim\Diamond p \strictif \sim\Diamond\sim\Diamond p$ (1)
[11.03] (1)(32.3 $\sim\Diamond p/p) \cdot = \cdot$ QED.

32.32 $\sim\Diamond\Diamond p \strictif \sim\Diamond\sim\Diamond\sim\Diamond\Diamond p$ [32.3 $\Diamond p/p$]

32.33 $\sim\Diamond\sim\Diamond\sim\Diamond\sim\Diamond\Diamond p = \sim\Diamond\sim\Diamond\Diamond p$ [32.31 $\Diamond p/p$]

32.4 $p \strictif q \cdot \strictif \cdot \sim\Diamond\Diamond q \strictif \sim\Diamond\Diamond p$
[11.6] (31.1)(30.1 $\Diamond p/p, \Diamond q/q) \strictif$ QED.

32.41 $p \strictif q \cdot \strictif \cdot \sim\Diamond\sim\Diamond p \strictif \sim\Diamond\sim\Diamond q$
[11.6] (30.1)(30.1 $\sim\Diamond q/p, \sim\Diamond p/q) \strictif$ QED.

32.42 $\sim\Diamond p \prec \sim\Diamond q \,.\prec .\, \sim\Diamond\Diamond p \prec \sim\Diamond\Diamond q$
[12.3] $(31.11\ \sim\Diamond p/p,\ \sim\Diamond q/q) =$ QED.

32.43 $\sim\Diamond\Diamond p \prec \sim\Diamond q \,.\prec .\, \sim\Diamond\Diamond p \prec \sim\Diamond\Diamond q$
[32.23] $(32.42\ \Diamond p/p) =$ QED.

32.5 $\sim\Diamond p \prec \sim\Diamond\sim\Diamond\Diamond\sim\Diamond\Diamond p$
[19.74] $\sim\Diamond p \prec .\, p \prec q.$
[32.4] $\prec .\, \sim\Diamond\Diamond q \prec \sim\Diamond\Diamond p.$
[32.4] $\prec .\, \sim\Diamond\Diamond\sim\Diamond\Diamond p \prec \sim\Diamond\Diamond\sim\Diamond\Diamond q.$
[22.11] $\prec .\, \sim\Diamond\Diamond\sim\Diamond\Diamond p \prec \Diamond\Diamond q$ (1)
[18.14] $\{(1)\ \sim\Diamond\Diamond p/q\} =$ QED.

32.51 $\sim\Diamond\sim\Diamond\sim\Diamond p \prec \sim\Diamond\sim\Diamond\Diamond\sim\Diamond\Diamond p$
[32.41] $(32.5) \prec :\, \sim\Diamond\sim\Diamond\sim\Diamond p \prec \sim\Diamond\sim\Diamond\sim\Diamond\sim\Diamond\Diamond\sim\Diamond\Diamond p$ (1)
[32.33] (1) = QED.

We now introduce a convenient method of abbreviation. We shall replace a single "$\Diamond$" by "1", "$\Diamond\Diamond$" by "2", and "$\sim$" by "-". All strings of more than two successive "$\Diamond$" reduce to two, by 32.23; and of course two successive "$\sim$" vanish. We shall devote our attention primarily to modalities in which these reductions have been made, and which are of the form $\sim \cdots \Diamond p$; (we shall call them simplified modalities of type A). They may be thought of as composed of "-1" and "-2".

The principles we have proved in 22.4–.65 and 32.3–.51 may be summed up and abbreviated as follows:

32.31 $\text{-1-1-1-1}p = \text{-1-1}p$ 32.33 $\text{-1-1-1-2}p = \text{-1-2}p$

32.4 $p \prec q \,.\prec .\, \text{-2}q \prec \text{-2}p$ 32.41 $p \prec q \,.\prec .\, \text{-1-1}p \prec \text{-1-1}q$

32.42 $\text{-1}p \prec \text{-1}q \,.\prec .\, \text{-2}p \prec \text{-2}q$ 32.43 $\text{-2}p \prec \text{-1}q \,.\prec .\, \text{-2}p \prec \text{-2}q$

32.6 $\text{-2}p \prec \text{-1}p \prec \text{-1-1-1}p \prec \text{-1-2-2}p \prec \text{-1-2-1}p$
[22.4, 32.3, 32.51, 22.65]

32.61 $\text{-2}p \prec \text{-1-1-2}p \prec \text{-1-1-1}p \prec \text{-1-2-2}p \prec \text{-1-2-1}p$
[32.32, 22.55, 32.51, 22.65]

32.62 $\text{-2-1}p \prec \text{-1-1}p \prec \text{-1-2}p$ [22.41.5]

32.63 $\text{-2-1}p \prec \text{-2-2}p \prec \text{-1-2}p$ [22.6.42]

Subsequent proofs will be in this abbreviated notation.

32.7 $\text{-2}p \prec \text{-2-2-2}p$ [32.42] $(32.5) \prec$ QED.

32.71 $\text{-2-2-2-2}p = \text{-2-2}p$ [32.4] $(32.7) \prec :\text{-2-2-2-2}p \prec \text{-2-2}p$ (1)
[11.03] $(1)(32.7\ \text{-2}p/p) =$ QED.

32.8 $\text{-2}p \prec \text{-2-1-2}p$ [32.43] $(32.32) \prec$ QED.

32.81 $\text{-2-1-2-1}p = \text{-2-1}p$ [32.6] $\text{-1}p \prec \text{-1-2-1}p$ (1)
[32.4] $(1) \prec :\text{-2-1-2-1}p \prec \text{-2-1}p$ (2)
[11.03] $(2)(32.8\ \text{-1}p/p) =$ QED.

32.82 -1-2-1-2p = -1-2p [30.1] (32.8) ⥽ : -1-2-1-2p ⥽ -1-2p (1)
[32.6 -2p/p] -1-2p ⥽ -1-2-1-2p (2)
[11.03] (1)(2) = QED.

32.83 -2-1-2-2p = -2-2p [32.6] -2p ⥽ -1-2-2p (1)
[32.4] (1) ⥽ : -2-1-2-2p ⥽ -2-2p (2)
[11.03] (2)(32.8 -2p/p) = QED.

32.84 -1-1-2-1-2p = -1-1-2p [32.82]

32.89 -2-1p ⥽ -1-1-2-1p ⥽ -1-1p ⥽ -1-2p
[32.8, 22.41] -2-1p ⥽ -2-1-2-1p ⥽ -1-1-2-1p (1)
[32.31] (22.55 -1p/p) = : -1-1-2-1p ⥽ -1-1p (2)
[(1)(2)(22.5)] QED.

32.9 -2-2p = -2-1p [32.4] (32.5) ⥽ : -2-1-2-2p ⥽ -2-1p (1)
[32.83] (1) = : -2-2p ⥽ -2-1p (2)
[11.03] (2)(22.6) = QED.

32.91 -2-1-2p = -2-2-2p = -2-2-1p = -2-1-1p [32.9]

32.92 -2-1-2-1p = -2-2-2-1p = -2-2-1-1p = -2-1-1-1p = -2-1p
[(32.91 -1p/p), 32.81]

32.93 -2-1-2-2p = -2-2-2-2p = -2-2-1-2p = -2-1-1-2p = -2-2p = -2-1p
[(32.91 -2p/p), 32.83.9]

32.95 -1-2-2p = -1-2-1p [32.9]

32.96 -1-1-2-2p = -1-1-2-1p [32.9]

32.97 -1-2-1-1p = -1-2p [32.91] -1-2-1-1p = -1-2-1-2p
[32.82] = -1-2p

32.98 -1-1-2-1-1p = -1-1-2p [32.97]

32.99 -2p ⥽ -2-1-1p ⥽ -1-1-2p ⥽ -1-1-1p ⥽ -1-2-1p
[32.8, (22.41 -2p/p), 32.61] -2p ⥽ -2-1-2p ⥽ -1-1-2p ⥽ -1-1-1p ⥽ -1-2-1p (1)
[32.91] (1) = QED.

The theorems in number 31—especially 31.3.4.5.51—provide a method for transforming principles of material implication into principles of strict implication. The generality of this method will be appreciated in view of the fact that, whenever $P \supset Q$ is a valid principle of material implication (i.e., of the ordinary two-valued calculus of propositions) P ⥽ Q can be proved in Lewis's system (even in S1).[9] However, this form of the method holds only for S3 or systems containing it, not for S2.

[9] Cf. M. Wajsberg, *Ein erweiterter Klassenkalkül*, ***Monatshefte für Mathematik und Physik***, vol. 40 (1933), pp. 113–126, especially pp. 115f. and Sätze 1–3, pp. 118f. For the proof of Satz 1.6, 18.53 p⥽q.$\sim\Diamond\sim p$:⥽.$\sim\Diamond\sim q$ may be used instead of Wajsberg's Axiom *e* (corresponding to our 31.11).

IV. Reduction to 42 modalities. We are chiefly concerned here with the reduction of modalities. The theorems just proved suffice to reduce the distinct modalities in S3 to a finite number. The key theorems are 32.1 (or 32.2), 32.3, and 32.5; all of these are independent of S2, as shown by Huntington's Example 0.5 (p. 28). We first introduce some *definitions.*[10]

Modal functions are defined thus: (1) The propositional variables $p, q, r, \cdots$ are modal functions; (2) If P and Q are modal functions, so are $\sim P$, $\Diamond P$, $P.Q$; (3) Any function equivalent by definition to a modal function is a modal function.

A *modality* is a modal function of one variable in whose construction no operators other than $\sim$ and $\Diamond$ are used. Note that some modal functions of one variable are not modalities; e.g., $\Diamond p . \Diamond \sim p$, $\sim p . \Diamond p$.

The *degree* of a modality equals the number of times its expression contains the sign "$\Diamond$".[11]

A *proper modality* is a modality of degree higher than zero.

The *type* of a proper modality is defined thus: If, after elimination of double negations (by $p = \sim\sim p$), a proper modality is of the form: (1) $\sim \cdots \Diamond p$, it is of *type A*, is an *A-modality*; (2) $\Diamond \cdots \Diamond p$ (or $\Diamond p$)—*type B*; (3) $\sim \cdots \Diamond \sim p$—*type C*; (4) $\Diamond \cdots \sim p$—*type D*.

An *affirmative modality* is one in which the number of times negation ($\sim$) is used is even; a *negative* modality is one in which the number is odd.

By 32.23 ($\Diamond\Diamond\Diamond p = \Diamond\Diamond p$) and 12.3 ($p = \sim\sim p$), any modality in S3 reduces to one whose expression contains no two negation-signs in immediate succession, and no more than two possibility-signs in immediate succession. We need consider only the modalities in which these reductions have been made, which we call *simplified modalities.*

Let us first consider modalities of type A. A-modalities reduce to those composed of the combinations -1 or -2 ($\sim\Diamond$ or $\sim\Diamond\Diamond$) in succession or alternation. There are three cases to be considered.

Case 1. The modalities composed of repetitions of -1 reduce to three forms: -1p, -1-1p, -1-1-1p. For, by 32.31 (-1-1-1-1p = -1-1p), those with an even number (≥ 2) of -1 are equivalent to -1-1p; those with an odd number (≥ 3) of -1 are equivalent to -1-1-1p.

Case 2. The A-modalities of the form -2$\cdots p$ reduce to three: -2p, -2-1p, -2-1-1p. By 32.92.93, any simplified A-modality—hence any A-modality—of this form with two or four negations is equivalent to -2-1p; if this reduction holds for every modality of this form with n (≥ 4) negations, it holds for every such modality with $n+2$ negations; therefore every affirmative A-modality of the

[10] Cf. O. Becker, op. cit.; R. Feys, *Les Logiques nouvelles des modalités, Revue néoscolastique de philosophie,* vol. 40 (1937), pp. 517–553, vol. 41 (1938), pp. 217–252,—especially no. 21 (p. 532f); W. T. Parry, *Zum Lewisschen Aussagenkalkül, Ergebnisse eines mathematischen Kolloquiums,* Heft 4 (1933), pp. 15–16.

[11] More generally, the *degree of a modal function* is defined thus: (1) The propositional variables $p, q, r, \cdots$ are of zero degree. (2) If P is of degree n, then $\sim P$ is of degree n, and (3) $\Diamond P$ is of degree $n+1$. (4) If P is of degree n, and Q of degree m, where $n \geq m$, then the degree of $P.Q$ and of $Q.P$ is n. (5) Any function equivalent by definition to a function of degree n is also of degree n.

form -2···*p* is equivalent to -2-1*p*. It follows that every negative A-modality of the form -2···*p* with three or more negations is equivalent to -2-1-1*p* or -2-1-2*p*, hence in any case (by 32.91) to -2-1-1*p*.

Case 3. The A-modalities of the form -1····-2···*p* reduce to four forms: -1-2*p*, -1-1-2*p*, -1-2-1*p*, -1-1-2-1*p*. By 32.33 (-1-1-1-2*p*=-1-2*p*), an odd number of -1 before the first -2 reduces to one -1; and any even number (≥ 2) of -1 reduces to two. By case 2 above, the part of the modality beginning with the first -2 (on the left) reduces to -2*p*, -2-1*p*, or -2-1-1*p*. This leaves six possible forms; but, by 32.97, -1-2-1-1*p*=-1-2*p*; and, by 32.98, -1-1-2-1-1*p*=-1-1-2*p*.

The A-modalities, then, reduce to ten forms: -1*p*; -1-1*p*, -1-2*p*; -1-1-1*p*, -1-1-2*p*, -1-2-1*p*; -1-1-2-1*p*; -2*p*, -2-1*p*, -2-1-1*p*.[12] These can be arranged according to the implications stated in 32.89 (affirmative modalities), 32.6.99 (negative); see table below.

By the principles 22.7 ($p=q.=.\sim p=\sim q$) and 12.44 ($p \strictif q.=.\sim q \strictif \sim p$), all equivalences and implications for type A modalities are transformed into corresponding equivalences and implications for type B modalities, which therefore likewise reduce to ten forms. Substituting -*p* for *p* in the principles for types A and B, we secure the principles for types C and D respectively. Thereby each of these types also reduces to ten forms. These reductions are easily reproduced. We limit ourselves, therefore, to presenting a summary table of these forty proper modalities, classified according to distinction of type and affirmative-negative, in the order of the implications which hold. A line is added for the two modalities of zero degree, *p* and -*p*, with implications as in S1.

Type *Affirmative* *Negative*

A: $\text{-2-1}p \strictif \text{-1-1-2-1}p \strictif \text{-1-1}p \strictif \text{-1-2}p;$

$$\text{-2}p \strictif \left\{\begin{array}{l}\text{-2-1-1}p \strictif \text{-1-1-2}p\\ \text{-1}p\end{array}\right\} \strictif \text{-1-1-1}p \strictif \text{-1-2-1}p.$$

B: $\text{1-2-1}p \strictif \text{1-1-1}p \strictif \left\{\begin{array}{l}\text{1-1-2}p \strictif \text{2-1-1}p\\ \text{1}p\end{array}\right\} \strictif \text{2}p;$

$$\text{1-2}p \strictif \text{1-1}p \strictif \text{1-1-2-1}p \strictif \text{2-1}p.$$

C: $\text{-2-}p \strictif \left\{\begin{array}{l}\text{-2-1-1-}p \strictif \text{-1-1-2-}p\\ \text{-1-}p\end{array}\right\} \strictif \text{-1-1-1-}p \strictif \text{-1-2-1-}p;$

$$\text{-2-1-}p \strictif \text{-1-1-2-1-}p \strictif \text{-1-1-}p \strictif \text{-1-2-}p.$$

D: $\text{1-2-}p \strictif \text{1-1-}p \strictif \text{1-1-2-1-}p \strictif \text{2-1-}p;$

$$\text{1-2-1-}p \strictif \text{1-1-1-}p \strictif \left\{\begin{array}{l}\text{1-1-2-}p \strictif \text{2-1-1-}p\\ \text{1-}p\end{array}\right\} \strictif \text{2-}p.$$

$$(\text{-2-}p \strictif \text{-1-}p \strictif\ )\ p\ (\ \strictif \text{1}p \strictif \text{2}p);\quad (\text{-2}p \strictif \text{-1}p \strictif\ )\ \text{-}p\ (\ \strictif \text{1-}p \strictif \text{2-}p).$$

Thus the modalities in S3 have been reduced to 42, none of which is higher than the fifth degree.

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[12] This reduction may be checked by constructing a "tree" of A-modalities, substituting alternately -1*p*, -2*p* for *p*, in *p* and in every resultant of this process, ending a "branch" at each modality which reduces to one of lower degree. The branches end at the following modalities: -1-2-2*p*; -1-1-1-1*p*, -1-1-1-2*p*, -1-1-2-2*p*, -1-2-1-1*p*, -1-2-1-2*p*; -1-1-2-1-1*p*, -1-1-2-1-2*p*; -2-2*p*; -2-1-2*p*; -2-1-1-1*p*, -2-1-1-2*p*, each of which reduces to one of the ten given above, by: 32.95; 32.31.33.96.97.82; 32.98.84; 32.9; 32.91; 32.92.93, respectively.

It will be observed that such transformations of principles for modalities of one type into corresponding principles for modalities of other types (by 22.7, 12.44, and substitution) hold for any system including S1. Therefore, in any system which includes S1 and in which no equivalence of modalities of different types holds, there is an equal number of irreducible modalities of each of the four types (in fact, an equal number for each degree of modality above zero). The total number of distinct modalities, then, in any such system must be of the form $4n+2$. In consistent systems containing equivalences between modalities of different types, the number of distinct modalities is sometimes of this form, as in the case of S5, sometimes of the form $4n$, (see below, VI, 6).

V. Irreducibility of the 42 modalities. The question naturally arises as to whether any further reductions are possible in S3—i.e., whether it is possible to establish any equivalences among the 42 modalities. It can be shown that, as a matter of fact, these 42 modalities are all distinct. This is done by means of four sets of matrices: Groups I, II, and III, in *Symbolic logic*, Appendix II,—which shall be referred to as Examples 0.1, 0.2, 0.3, respectively, following Huntington, §11; and a new example, called Example 0.7. Each of these examples satisfies the postulates of S3.

Ex. 0.7 is as follows: $K=(1, 2, 3, 4, 5, 6, 7, 8)$. T (the designated values) $=(1, 2)$. $p.q$ and $\sim p$ are defined as in Huntington's Ex. 0.4[13] (i.e., as ab and a' in Ex. 0.4); $\Diamond p$ (and $\sim\Diamond p$, for comparison with Huntington's a^*,) defined thus:

p	1	2	3	4	5	6	7	8
$\Diamond p$	1	1	3	3	1	1	3	7
$\sim\Diamond p$	8	8	6	6	8	8	6	2

The table for p⥽q will be the same as in Ex. 0.4, except that, in each case, (1, 2, 3, 4)⥽(5, 6) and (3, 4)⥽(7, 8) will be 6 (instead of 8).

1. Ex. 0.1 shows that the following implications fail: -1-2-1p⥽-1-1-1p, (-1-1-2p, -2-1-1p, -1p, -2p); -1-1-1p⥽-1-1-2p, (-2-1-1p, -2p); -1p⥽-1-1-2p, (-2-1-1p, -2p); -1-2p⥽-1-1-1p, (-1-1-2-1p, -2-1p); -1-1p⥽-1-1-2-1p, (-2-1p). Ex. 0.2 shows that the following fail: -1-1-1p⥽-1p; -1-1-2p⥽-1p, (-2p); -2-1-1p⥽-1p, (-2p). Ex. 0.7 shows that the following fail: -1-1-2p⥽-2-1-1p; -1-1-2-1p⥽-2-1p. Ex. 0.3 (which satisfies the postulates of S5, reducing all modalities to the first degree,) shows that no affirmative modality implies any negative modality, and vice versa. This accounts for all possible implications—let alone equivalences—among A-modalities, other than those summarized in the table of implications.

[13] Ex. 0.4 is the writer's example in *The postulates for "strict implication"*, *Mind*, vol. 43 (1934), pp. 78–80, introduced to distinguish S2 from S3, with different designation of the elements.

The definition may be expressed thus: Let the elements be understood as designating classes. 1 = the universe class, composed of the mutually exclusive classes 4, 6, and 7. The complements of 1, 4, 6, 7 are designated by 8, 5, 3, 2 respectively. Then $p.q$ = the logical product of p and q; and $\sim p$ = the complement of p (making the system a Boolean algebra).

These same examples give the same results for the corresponding modalities of the other three types, (using, again, 22.7, 12.44, and substitution of -p for p). Hence no implications hold among the modalities of any given type, other than those summarized in the table of implications.

2. It is readily shown that no proper modality of one type is equivalent in S3 to any modality of another type. (a) Consider, first, the fact that in S5 the affirmative modalities of types A and B reduce to $\Diamond p$; the affirmatives of types C and D reduce to $\sim\Diamond\sim p$; the negatives of types A and B reduce to $\sim\Diamond p$; and the negatives of types C and D, to $\Diamond\sim p$.[14] Consequently, Ex. 0.3 (which satisfies S5 as well as S3) shows that no affirmative modality of type A or B implies any affirmative of type C or D; and no negative of type C or D implies any negative of type A or B. (b) Ex. 0.3 likewise shows that no affirmative modality of any type implies any negative modality of any (other) type, and no negative implies any affirmative.

(c) Ex. 0.1 shows that the following implications fail: 1-2-1p $\prec$ -1-2p, 1-2p $\prec$ -1-2-1p fail—hence no type B affirmative implies any type A affirmative, and no type B negative implies any type A negative; 1-2-p $\prec$ -1-2-1-p, 1-2-1-p $\prec$ -1-2-p fail—hence no type D affirmative implies any type C affirmative, and no type D negative implies any type C negative; 1-2-p $\prec$ -1-2p, 1-2p $\prec$ -1-2-p fail—hence no type D affirmative implies any type A affirmative, no type B negative implies any type C negative.

The only possible inter-type implications, then, are the following (and such actually exist): certain affirmative modalities of type C imply certain affirmatives of types A, D, and B; certain affirmatives of types A and D imply certain affirmatives of type B; certain negatives of type A imply certain negatives of types B, C, and D; and certain negatives of types B and C imply certain negatives of type D. Therefore no proper modality of one type is equivalent to any modality of another type.

3. There remains only the question of the zero-degree modalities, p and -p. Ex. 0.1 shows that p does not imply any type A affirmative. Ex. 0.3 shows that no type A affirmative implies p, and that -p does not imply any type A negative; —also that no implication holds between p and any negative modality, nor between -p and any affirmative. Ex. 0.2 shows that -2-1-1p $\prec$ -p fails. The implications -2p $\prec$ -p, -1p $\prec$ -p are therefore the only ones holding between A-modalities and zero-degree modalities. Similarly, applying these same examples to the corresponding questions with regard to the three other types of modality, we conclude that the only implications between proper modalities and zero-degree modalities are those summarized in the line:

$$-2\text{-}p \prec -1\text{-}p \prec p \prec 1p \prec 2p; \qquad -2p \prec -1p \prec -p \prec 1\text{-}p \prec 2\text{-}p.$$

We have shown, then, that no possible equivalence exists among these 42 modalities in S3. The modalities of S3 therefore reduce to 42 distinct and irreducible modalities. (This, it will be recalled, does not account for the modal functions of one variable involving conjunction.)

[14] Cf. Becker, op. cit., I, §2.

It is readily ascertained that the number of possibly distinct modalities of the nth degree in any form of Lewis's system is 2^{n+1} S3 contains the following number of distinct and irreducible modalities, by degree: 0 degree: 2 (out of 2 possible); 1st degree: 4 (out of 4 possible); 2d degree: 8 (out of 8 possible); 3d degree: 12 (out of 16 possible); 4th degree: 12 (out of 32 possible); 5th degree: 4 (out of 64 possible); nth degree ($n \geq 6$): 0 (out of 2^{n+1} possible).

VI. Systems including S3. 1. S4. From the theorem 32.1 $\sim\Diamond\sim\sim\Diamond\sim q \prec . \sim\Diamond\sim p \prec \sim\Diamond\sim\sim\Diamond\sim p$, it follows immediately that adding to S3 any principle of the form $\sim\Diamond\sim\sim\Diamond\sim P$ will give as a theorem

C10 $\quad \sim\Diamond\sim p \prec \sim\Diamond\sim\sim\Diamond\sim p,$

the characteristic principle of S4.

Hence, the postulates for S3, together with

41.1 $\quad \sim\Diamond\sim(p \prec p),$

which is equivalent to $\sim\Diamond\sim\sim\Diamond\sim(p \supset p)$ by 18.7, constitute a postulate set for S4.

If we add to S3 the rule of procedure:

41.2 If P has been asserted, $\sim\Diamond\sim P$ may be asserted,

we can obviously prove 41.1, and hence C10. This is 25.2 in Feys, op. cit.—borrowed from an axiomatization by K. Gödel for a system equivalent to S4. It follows that Feys' "logique t," (op. cit., 28.1,) which includes this rule 25.2, is *not* equivalent to S3, as he seems to think (op. cit., 43.2). If 30.1 (A8) is a theorem in "logique t," this system is equivalent to S4 (Feys' "logique r"); if 30.1 is not a theorem, then "logique t" neither includes nor is included in S3.

We take this occasion to give the proof that, if we replace 30.1 (A8) in our postulate set for S3 by C10 $\sim\Diamond\sim p \prec \sim\Diamond\sim\sim\Diamond\sim p$, then 19.01 (B8) and 30.1 follow as theorems.[15] It will be recalled that our postulates without 30.1 are a postulate set for S1.

First the lemma:

41.3 $\quad p \prec q . = . \sim\Diamond\sim(p \prec q)$

[11.03] $\quad$ (C10 $p \supset q/p$)(18.42 $\sim\Diamond\sim(p \supset q)/p$) $= :$

$\sim\Diamond\sim(p \supset q) = \sim\Diamond\sim\sim\Diamond\sim(p \supset q)$ (1)

[18.7] $\quad$ (1) = QED.

(B8) $\quad \Diamond(pq) \prec \Diamond p$

[41.3] $\quad$ (11.2) $= : \sim\Diamond\sim(pq \prec p)$ (1)

[18.53] $\quad$ (15.2 $pq \prec p/p, r/q$)(1) $\prec : \sim\Diamond\sim(r \supset . pq \prec p)$ (2)

[18.7] $\quad$ (2) $= : r \prec . pq \prec p$ (3)

[16.33] $\quad \{(3)\ \Diamond(pq)/r\} = : \Diamond(pq) \prec . \Diamond(pq) . pq \prec p .$

[11.1] $\quad \prec . pq \prec p . \Diamond(pq) .$

[18.51] $\quad \prec . \Diamond p$ QED.

[15] This fact is mentioned but not proved in *Symbolic logic*, p. 501.

Hence follow the theorems of S2 (22.3 in particular).

(A8) $p \mathrel{-\!\!3} q \,.\, \mathrel{-\!\!3} \,.\, \sim\Diamond q \mathrel{-\!\!3} \sim\Diamond p$

[14.26] (18.5) $= : p \mathrel{-\!\!3} q \,.\, \mathrel{-\!\!3} \,.\, \sim\Diamond q \supset \sim\Diamond p$ (1)

[12.44] (1) $= : \sim(\sim\Diamond q \supset \sim\Diamond p) \,.\, \mathrel{-\!\!3} \,.\, \sim(p \mathrel{-\!\!3} q)$ (2)

[22.3, (2)] $\sim\Diamond\sim(p \mathrel{-\!\!3} q) \mathrel{-\!\!3} \sim\Diamond\sim(\sim\Diamond q \supset \sim\Diamond p)$ (3)

[41.3, 18.7] (3) = QED.

By the corollary of C10, $\Diamond\Diamond p = \Diamond p$, all modalities in S4 reduce to alternations of $\sim$ and $\Diamond$; and by 32.31, $\sim\Diamond\sim\Diamond\sim\Diamond\sim\Diamond p = \sim\Diamond\sim\Diamond p$, such alternations reduce to those of the third degree at highest. There remain twelve proper modalities, together with p and $\sim p$, for which the following implications hold (by principles of S3):

$$\sim\Diamond\sim p \mathrel{-\!\!3} \sim\Diamond\sim\Diamond\sim\Diamond\sim p \mathrel{-\!\!3} \begin{Bmatrix} \Diamond\sim\Diamond\sim p \\ \sim\Diamond\sim\Diamond p \end{Bmatrix} \mathrel{-\!\!3} \Diamond\sim\Diamond\sim\Diamond p \mathrel{-\!\!3} \Diamond p;$$

$$\sim\Diamond p \mathrel{-\!\!3} \sim\Diamond\sim\Diamond\sim\Diamond p \mathrel{-\!\!3} \begin{Bmatrix} \Diamond\sim\Diamond p \\ \sim\Diamond\sim\Diamond\sim p \end{Bmatrix} \mathrel{-\!\!3} \Diamond\sim\Diamond\sim\Diamond\sim p \mathrel{-\!\!3} \Diamond\sim p.$$

$$\sim\Diamond\sim p \mathrel{-\!\!3} p \mathrel{-\!\!3} \Diamond p; \qquad \sim\Diamond p \mathrel{-\!\!3} \sim p \mathrel{-\!\!3} \Diamond\sim p.$$

That these 14 modalities can not be further reduced is shown by Example 0.3 (Group III), together with a new example for S4 which we call Example 0.8.

Example 0.8 is defined as follows: K, T, $p.q$, and $\sim p$ as in Ex. 0.4; $\Diamond p$ (and $\sim\Diamond p$) as in this table:

p	1	2	3	4	5	6	7	8
$\Diamond p$	1	2	3	4	1	2	3	8
$\sim\Diamond p$	8	7	6	5	8	7	6	1

Ex. 0.8 shows that the following implications fail:

$$\Diamond\sim\Diamond\sim p \mathrel{-\!\!3} \sim\Diamond\sim\Diamond p; \quad \sim\Diamond\sim\Diamond\sim\Diamond\sim p \mathrel{-\!\!3} p; \quad p \mathrel{-\!\!3} \Diamond\sim\Diamond\sim\Diamond p.$$

Ex. 0.3 shows that $\sim\Diamond\sim\Diamond p \mathrel{-\!\!3} \Diamond\sim\Diamond\sim p$ fails; also that no affirmative modality implies or is implied by any negative. These examples give the same results for the corresponding implications for the negative modalities. From these independences, in conjunction with the implications summarized in the table of implications, it follows that no implication holds among the modalities of S4 other than those in the table.

2. Systems intermediate between S3 and S4 are readily constructed by adding to S3 some implication between modalities which holds in S4 but not in S3. For example, adding to S3

42.1 $\sim\Diamond\sim\Diamond\Diamond p \mathrel{-\!\!3} \sim\Diamond\sim\Diamond p$

gives the theorem -1-2p=-1-1p, and hence we have

$$\text{-1-1-2}p = \text{-1-1-1}p = \text{-1-2-1}p, \qquad \text{-1-1-2-1}p = \text{-1-1}p = \text{-1-2}p.$$

The table for the (24+2) modalities remaining is as follows:

Type	*Affirmative*	*Negative*
A:	$\text{-2-1}p \prec \text{-1-1}p;$	$\text{-2}p \prec \left\{\begin{matrix}\text{-2-1-1}p \\ \text{-1}p\end{matrix}\right\} \prec \text{-1-1-1}p.$
B:	$\text{1-1-1}p \prec \left\{\begin{matrix}\text{2-1-1}p \\ \text{1}p\end{matrix}\right\} \prec \text{2}p;$	$\text{1-1}p \prec \text{2-1}p.$
C:	$\text{-2-}p \prec \left\{\begin{matrix}\text{-2-1-1-}p \\ \text{-1-}p\end{matrix}\right\} \prec \text{-1-1-1-}p;$	$\text{-2-1-}p \prec \text{-1-1-}p.$
D:	$\text{1-1-}p \prec \text{2-1-}p;$	$\text{1-1-1-}p \prec \left\{\begin{matrix}\text{2-1-1-}p \\ \text{1-}p\end{matrix}\right\} \prec \text{2-}p.$
	$(\text{-2-}p \prec \text{-1-}p \prec)\ p\ (\prec \text{1}p \prec \text{2}p);$	$(\text{-2}p \prec \text{-1}p \prec)\ \text{-}p\ (\prec \text{1-}p \prec \text{2-}p).$

That these 26 modalities are distinct in this system can be shown by means of Examples 0.3, 0.7, 0.8.

Ex. 0.8 shows that $\text{-2-1-1}p \prec \text{-1}p$ fails. Ex. 0.7 shows that $\text{-1}p \prec \text{-2-1-1}p$, $\text{-1-1}p \prec \text{-2-1}p$ fail. These same examples show that the corresponding implications for the other types fail. It follows that no implication holds between two affirmative (or negative) modalities of any given type other than those indicated in the above table. Examples 0.3 and 0.8 show that there is no equivalence between modalities of different types,[16] that there is no implication between an affirmative and a negative modality, and that there is no implication between a zero-degree and a proper modality other than those summarized in the last line of the table; for, in each of these three sorts of cases, there is no such equivalence or implication in S4.

3. S4.5. Becker effected a reduction of modalities to ten by means of C10 $\sim\Diamond\sim p \prec \sim\Diamond\sim\sim\Diamond\sim p$ and C12 $p \prec \sim\Diamond\sim\Diamond p$.[17] But it turns out that then C11 $\Diamond p \prec \sim\Diamond\sim\Diamond p$ is a theorem, and the system reduces to S5, with only six irreducible modalities.[18] However, if we add to S4 (i.e., S1 plus C10) the principle

51.1 $\quad \sim\Diamond\sim\Diamond\sim\Diamond p \prec \sim\Diamond p,$

(which follows from C12 by 22.3) we obtain all the reductions of Becker's so-called "ten-modalities calculus." For we already have the converse, 32.3, and hence $\sim\Diamond\sim\Diamond\sim\Diamond p = \sim\Diamond p$, $\Diamond\sim\Diamond\sim\Diamond p = \Diamond p$, $\sim\Diamond\sim\Diamond\sim\Diamond\sim p = \sim\Diamond\sim p$, and $\Diamond\sim\Diamond\sim\Diamond\sim p = \Diamond\sim p$. The modalities of this system—which we call S4.5—then reduce to the following (compare table for S4):

$$\sim\Diamond\sim p \prec \left\{\begin{matrix}\Diamond\sim\Diamond\sim p \\ \sim\Diamond\sim\Diamond p\end{matrix}\right\} \prec \Diamond p; \qquad \sim\Diamond p \prec \left\{\begin{matrix}\Diamond\sim\Diamond p \\ \sim\Diamond\sim\Diamond\sim p\end{matrix}\right\} \prec \Diamond\sim p.$$

$$\sim\Diamond\sim p \prec p \prec \Diamond p; \qquad \sim\Diamond p \prec \sim p \prec \Diamond\sim p.$$

That 51.1 is independent of S4 is shown by Ex. 0.2. However, we have no proof that S4.5 does not reduce to S5, (though there appears to be no way

[16] There are implications, however. For example, any affirmative (or negative) modality of type A implies any affirmative (or negative, respectively) of type B; any affirmative (or negative) of type C implies any affirmative (or negative, respectively) of type D.

[17] Op. cit., §3 (pp. 512–517).

[18] ***Symbolic logic***, p. 498. The proof of this fact, by the way, was first pointed out to me by Mr. Y. T. Shen.

to equate modalities of different types in S4.5). If an example to show this distinction exists, it must have at least sixteen elements.

S4.5 could also be obtained by adding to S4 the weaker principle

51.2 $\sim\Diamond\sim\Diamond\sim\Diamond p \prec \sim p$,

or the equivalent, $p \prec \Diamond\sim\Diamond\sim\Diamond p$. For, by the principle of S4 $\Diamond\Diamond p = \Diamond p$, $(51.2\ \Diamond p/p) = 51.1$.

4. The system formed by adding the above principle 51.1 $\sim\Diamond\sim\Diamond\sim\Diamond p \prec \sim\Diamond p$ to S3 is worth noticing. As before, this gives us -1-1-1p=-1p; from which follow -1-1-2p=-2p, -1-1-2-1p=-2-1p. Then, since -2$p\prec$-2-1-1$p\prec$-1-1-2p, -2-1-1p=-2p. The table of implications then reduces to the following:

A:	-2-1p $\prec$ -1-1p $\prec$ -1-2p;	-2p $\prec$ -1p $\prec$ -1-2-1p.
B:	1-2-1p $\prec$ 1p $\prec$ 2p;	1-2p $\prec$ 1-1p $\prec$ 2-1p.
C:	-2-p $\prec$ -1-p $\prec$ -1-2-1-p;	-2-1-p $\prec$ -1-1-p $\prec$ -1-2-p.
D:	1-2-p $\prec$ 1-1-p $\prec$ 2-1-p;	1-2-1-p $\prec$ 1-p $\prec$ 2-p.
	(-2-p $\prec$ -1-p $\prec$) p ($\prec$ 1p $\prec$ 2p);	(-2p $\prec$ -1p $\prec$) -p ($\prec$ 1-p $\prec$ 2-p).

These 26 modalities are distinct in this system, as is shown by Examples 0.1 and 0.3. Ex. 0.1 shows that the converse of each of the above implications fails. Ex. 0.1 and 0.3 show that no modality is equivalent to any modality of a different type, and that no affirmative modality implies or is implied by a negative, in exactly the same way as they show this for S3. Ex. 0.1 shows that -$p\prec$-1-2-1p, -1-2-1$p\prec$-p, $p\prec$-1-2p, all fail, and Ex. 0.3 shows that -2-1$p\prec p$ fails; hence no implication holds between zero-degree modalities and A-modalities, other than -2$p\prec$-1$p\prec$-p. Similarly for the corresponding implications between zero-degree modalities and B, C, and D modalities.

If this system is compared with the system above (under 2.) consisting of S3 plus 42.1 (-1-2$p\prec$-1-1p), which also had just 26 distinct modalities, it is found that the present system neither includes the earlier one (by Ex. 0.1), nor is included in it (by Ex. 0.2). Likewise, this system neither includes nor is included in S4 (by the same examples). The *number* of distinct modalities, then, in a system including S3, does not uniquely determine the system. Also, the systems intermediate between S3 and S4.5 do not form a single series of decreasing strictness.

5. S5. Our theorems for S3 make it possible to show certain ways in which S5 can be obtained other than by adding C11 to S1.[19] (*a*) In the first place, S3 plus

C12 $p \prec \sim\Diamond\sim\Diamond p$

proves to be a postulate set for S5. We first prove C10:

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[19] Churchman (op. cit., pp. 78–81) has shown that S5 is obtained by adding to S2 the principle C15′ $p\prec\sim\Diamond\Diamond\sim\Diamond p$.

S5 is notable as reducing all modal functions to those of the first (or zero) degree—for which an Entscheidungsverfahren has been given: M. Wajsberg, op. cit.; W. T. Parry, *Zum Lewisschen Aussagenkalkül*, loc. cit.

C10 $\sim\Diamond\sim p \prec \sim\Diamond\sim\sim\Diamond\sim p$

[31.11]	C12 $\prec : \sim\Diamond\sim p \prec \sim\Diamond\sim\sim\Diamond\sim\Diamond p$	(1)
[18.7]	(12.1) $= : \sim\Diamond\sim(p \supset p)$	(2)
[(1)]	(2) $\prec : \sim\Diamond\sim\sim\Diamond\sim\Diamond(p \supset p)$	(3)
[32.1]	(3) $\prec$ QED.	

But C10 and C12 give C11, as noted above, so that the system reduces to S5.

The exact effect of adding C12 to S2 is not known as yet.

(*b*) Feys suggests four postulates which, when added to S2, reduce all modalities to those of the second degree.[20] If we add to S3 one of these, viz.

52.1 $\Diamond\Diamond p \prec \sim\Diamond\sim\Diamond\Diamond p,$

corresponding to Feys 29.44, (where (52.1) $=$ (C11 $\Diamond p/p$),) then C11 follows as theorem, and the system reduces to S5.[21]

C11 $\Diamond p \prec \sim\Diamond\sim\Diamond p$

[(18.42 $\Diamond\Diamond p/p$)(52.1)]	$\sim\Diamond\sim\Diamond\Diamond p = \Diamond\Diamond p$	(1)
[22.7, 12.3]	(1) $= : \Diamond\sim\Diamond\Diamond p = \sim\Diamond\Diamond p$	(2)
[(1)]	$\sim\Diamond\sim\Diamond\Diamond\sim\Diamond\Diamond p = \Diamond\Diamond\sim\Diamond\Diamond p$	
[(2)]	$= \Diamond\sim\Diamond\Diamond p$	
[(2)]	$= \sim\Diamond\Diamond p$	(3)
[(3)]	(32.5) $= : \sim\Diamond p \prec \sim\Diamond\Diamond p$	(4)
[(22.4)(4)]	$\sim\Diamond\Diamond p = \sim\Diamond p$	(5)
[22.7]	(5) $= : \Diamond\Diamond p = \Diamond p$	(6)
[(6)]	(52.1) $=$ QED.	

6. Each of the systems of strict implication we have considered up to this point is included in S5, and has $4n+2$ distinct modalities. There are systems, however, which have neither of these properties. (*a*) One such is obtained by adding to S4 (i.e., S1 plus C10) the principle

C16 $\Diamond\sim\Diamond\sim p = \sim\Diamond\sim\Diamond p.$

[20] Op. cit., 29.2–.4. Feys' postulates 29.41 (corresponding to our 32.2) and 29.42 in this "logique tt" are redundant.

The total number of possibly distinct modalities of degrees 0 through n is $(2+2^2+2^3+\cdots+2^{n+1}=)\ 2^{n+2}-2$. Feys' "logique tt" therefore has a maximum of 14 distinct modalities.

[21] Actually Feys states his principles with *material* implication (or equivalence) as the main relation; this can then be made into a strict relation by means of his rule of procedure 25.2—our 41.2.

If we add to S1 the postulates

52.2 $\Diamond\sim\Diamond\sim p \prec \sim\Diamond\sim\Diamond\sim\Diamond\sim p,$

corresponding to Feys 29.43, and 51.1 $\sim\Diamond\sim\Diamond\sim\Diamond p \prec \sim\Diamond p$, then C11 follows, thus:

[12.3]	(52.2 $\sim p/p$) $= : \Diamond\sim\Diamond p \prec \sim\Diamond\sim\Diamond\sim\Diamond p$	
	[51.1] $\prec \sim\Diamond p$	(1)
[12.45]	(1) $=$ C11.	

51.1 follows (by 22.3) from C12, which corresponds to Feys 28.31. Therefore Feys' "logique ss" (29.5), formed by adding Feys 28.31 to his "logique tt," reduces to his "logique à modalités simples," i.e., S5.

S4 of course contains C10a $\sim\Diamond\sim p = \sim\Diamond\sim\sim\Diamond\sim p$.
Therefore:

$$\begin{array}{llll} [C16] & \sim\Diamond\sim\Diamond\sim\Diamond\sim p & = \Diamond\sim\Diamond\sim\sim\Diamond\sim p & \\ [C10a] & & = \Diamond\sim\Diamond\sim p & \text{(i)} \\ [22.7, 12.3] & \{\text{(i)} \sim p/p\} & = : \Diamond\sim\Diamond\sim\Diamond p = \sim\Diamond\sim\Diamond p & \text{(ii)} \end{array}$$

Substituting $\sim p$ for p in C16, (i), and (ii) gives us (by 12.3) the corresponding principles for negative modalities, viz.,

$$\Diamond\sim\Diamond p = \sim\Diamond\sim\Diamond\sim p, \quad \sim\Diamond\sim\Diamond\sim\Diamond p = \Diamond\sim\Diamond p,$$

$$\Diamond\sim\Diamond\sim\Diamond\sim p = \sim\Diamond\sim\Diamond\sim p.$$

The modalities in this system therefore reduce to the following (cf. table of implications for S4, under 1. above):

$$\sim\Diamond\sim p \dashv \left\{\begin{matrix}\sim\Diamond\sim\Diamond p\\ p\end{matrix}\right\} \dashv \Diamond p; \qquad \sim\Diamond p \dashv \left\{\begin{matrix}\Diamond\sim\Diamond p\\ \sim p\end{matrix}\right\} \dashv \Diamond\sim p.$$

Ex. 0.2 shows that no further implications hold among these eight modalities.

This system neither includes S5 (by Ex. 0.2) nor is included in it (by Ex. 0.3).

If C11 were added, the system would reduce to material implication; for we would have:

$$\Diamond p = \sim\Diamond\sim\Diamond p = \Diamond\sim\Diamond\sim p = \sim\Diamond\sim p.$$

(*b*) A system that is not included in S5, and has $4n$ distinct modalities, can also be formed by adding the above principle C16 to S3. It can be proved that the modalities in this system reduce to the following:

$$\text{-2-}p \dashv \text{-1-}p \dashv \left\{\begin{matrix}\text{-1-1}p\\ p\end{matrix}\right\} \dashv 1p \dashv 2p; \qquad \text{-2}p \dashv \text{-1}p \dashv \left\{\begin{matrix}\text{1-1}p\\ \text{-}p\end{matrix}\right\} \dashv \text{1-}p \dashv \text{2-}p.$$

This system has 12 distinct modalities, unless C10 is a theorem—in which case it reduces to the previous system under (*a*), with eight modalities.

7. Becker suggested that the principle

$$1.9^* \qquad \sim\Diamond\sim\Diamond p \dashv \sim\Diamond\sim p$$

is a possible alternative to C11, which would bring about a similar reduction of modalities, though it has not as great an appeal to our logical intuition.[22] Churchman thinks that 1.9* and the corollary of this,

$$1.91^* \qquad \sim\Diamond\sim\Diamond p \dashv p,$$

the converse of C12, though "paradoxical," do not necessarily reduce the modalities to a finite number.[23]

As a matter of fact, in view of our theorem

$$22.8 \qquad \sim\Diamond\sim\Diamond(p \supset \sim\Diamond\sim p),$$

[22] Becker, op. cit., *Bemerkung zu* §2 (p. 516f.).
[23] Churchman, op. cit., pp. 78, 81n.

1.91*—and hence 1.9*—reduces not only S3 but S2 to the system of material implication, since it leads to the theorem $p = \sim\Diamond\sim p$, as follows:

[1.91*]	$(22.8) \prec : p \supset \sim\Diamond\sim p$	(1)
[14.29]	$(1) . \{(1)(p \supset \sim\Diamond\sim p)/p\} . \prec : \sim\Diamond\sim(p \supset \sim\Diamond\sim p)$	(2)
[18.7]	$(2) = : p \prec \sim\Diamond\sim p$	(3)
[(3)(18.42)]	$p = \sim\Diamond\sim p$	

—which reduces all modalities to the zero degree, and is therefore inconsistent with 20.01 (B9).

The (apparently) weaker principle $\sim\Diamond\sim\Diamond p \supset p$ would give the same result, (using 14.29 in the first line of the proof).

Churchman's C14 $\sim\Diamond\sim p = \sim\Diamond\sim\Diamond p$ would reduce even S1 to material implication. By C14 (and 12.3),

$$\sim\Diamond\sim p = \sim\Diamond\sim\Diamond p = \sim\Diamond\sim\Diamond\sim\sim p = \sim\Diamond\sim\Diamond\sim\Diamond\sim p = \sim\Diamond\sim\sim\Diamond\sim p;$$

22.8, a theorem in S2, is also a theorem in S4; and $p = \sim\Diamond\sim p$ would follow.

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